Comparative Evaluation of Llama Guard 3 and GPT OSS Safeguard as LLM Guardrails for a University Administrative Chatbot

Jan Walczak
Faculty of Electrical Engineering
Warsaw University of Technology
Warsaw, Poland
jan.walczak4.stud@pw.edu.pl

Bartosz Sawicki
Faculty of Electrical Engineering
Warsaw University of Technology
Warsaw, Poland
bartosz.sawicki@pw.edu.pl

2026 Progress in Applied Electrical Engineering (PAEE) | 979-8-3195-0938-3/26/$31.00 ©2026 IEEE | DOI: 10.1109/PAEE71887.2026.11660667

***Abstract*—Integrating large language models (LLMs) into academic infrastructure requires rigorous safety mechanisms. This paper presents a comparative evaluation of two guard models, Llama Guard 3 (with a Low-Rank Adaptation (LoRA) adapter) and GPT OSS Safeguard, in the context of securing an administrative chatbot at Warsaw University of Technology (WUT), incorporating a custom Topic Restriction category that enforces domain boundaries. Results reveal a fundamental trade-off: Llama Guard 3 exhibits high recall and low classification latency, but at the cost of numerous false positives. GPT OSS Safeguard achieves higher precision and better threat categorization, yet its response time precludes synchronous deployment. The study also revealed a significant discrepancy between expert labels and user perception, underscoring the inherent subjectivity of safety classification.**

***Index Terms*—Large Language Models, Guardrails, Chatbot Safety, Educational Technology, Content Moderation, Llama Guard, User Study**

I. Introduction

Conversational systems based on large language models are being more commonly adopted in higher education as administrative assistants. Such deployments face the problem of scope creep: systems designed for administrative queries are gradually repurposed for tasks they were not intended for (code generation, homework solving, essay writing), incurring unjustified costs, risk of hallucinating academic facts, and vulnerability to jailbreak attacks [1].

These challenges make it necessary to evaluate guardrail mechanisms capable not only of detecting toxicity, but also of effectively enforcing domain restrictions. The problem is further complicated by the language barrier – safety models are trained primarily on English-language datasets, and their effectiveness for under-represented languages such as Polish remains unclear [2].

This paper presents a comparative evaluation of Llama Guard 3 (open-source model with LoRA fine-tuning) and GPT OSS Safeguard (zero-shot model with Chain-of-Thought (CoT) reasoning) in the context of the Warsaw University of Technology administrative chatbot, incorporating the custom category S15 (Topic Restriction). The main research questions concern: (1) domain restriction effectiveness, (2) robustness against prompt injection attacks, (3) the over-refusal problem, and (4) the cost-effectiveness of open-source versus commercial models. Among the contributions is one of the first comparisons of these systems in the context of a strictly administrative university chatbot with a high-rigor topic restriction policy.

979-8-3195-0938-3/26/$31.00 © 2026 IEEE

The emergence of compound AI systems [3] has led to the establishment of external guardrail layers as a necessary component for supervising LLM input and output [4]. Built-in safety mechanisms based on Reinforcement Learning from Human Feedback (RLHF) fail due to the conflict between helpfulness and safety objectives, as well as generalization errors on unusual input formats [1]. Strict filters in turn amplify over-refusal, reaching up to 50% for safe queries containing ambiguous keywords [5]. The most serious threat vector in academic settings are prompt injection and jailbreak attacks, in which students deliberately attempt to circumvent safety filters using techniques ranging from role-play scenarios to intent obfuscation (Base64 encoding, ciphers) [1].

LLM integration in education also introduces scope creep: assistants designed for administrative support are often used for academic tasks, pushing them beyond their intended domain [6]. Over-reliance on AI for academic purposes risks eroding students' critical thinking skills [7]. Enforcing domain restrictions, however, creates tension between utility and safety [5].

Llama Guard 3 is a generative model (8B) based on the Llama 3 architecture [8], enabling safety policies to be defined as textual instructions [9]. It supports 14 threat categories from the MLCommons benchmark [10] and achieves competitive results in binary safety classification on Polish-language data after fine-tuning, though with a potential over-refusal risk [2]. GPT OSS Safeguard (20B) is a reasoning model with a native Chain-of-Thought mechanism, requiring no fine-tuning, where the policy is injected into the system prompt [11]. The model employs an instruction hierarchy, giving priority to system instructions over user requests [12].

In the Polish-language context, the dominance of English in LLM development leads to a significant quality gap [2].

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 16:09:55 UTC from IEEE Xplore. Restrictions apply.

Local models such as Bielik Guard [13] – a compact family of encoder-based classifiers (RoBERTa) – achieve high accuracy on Polish text classification (precision 77.65%, false positive rate (FPR) 0.63%), outperforming large multilingual models in this domain. However, a rigid classification head supporting exactly five predefined categories prevents taxonomy changes without costly retraining, the model by design omits jailbreak attack detection, and its exclusively Polish-language nature does not meet the requirements of an academic environment with a large international student population. For these reasons, it is necessary to turn to the newer generation of generative guard models capable of multilingual reasoning and flexible domain policy adaptation.

This paper constitutes one of the first comparative analyses of Llama Guard 3 and GPT OSS Safeguard in the context of a strictly administrative university chatbot with a custom Topic Restriction category. A further contribution is the creation of a Polish-language ground truth dataset of 1802 samples, collected during a 24-hour red teaming study involving over 30 Warsaw University of Technology (WUT) students – a dataset that, due to the lack of publicly available safety benchmarks for Polish educational chatbots, constitutes a standalone research contribution. The technical analysis is complemented by an empirical comparison of expert labels with subjective user perception, revealing discrepancies that are difficult to capture through classification metrics alone. The paper proceeds as follows: methodology (Section II), evaluation results (Section III), and conclusions (Section IV).

II. Methodology

Experiments were conducted on a workstation with an NVIDIA RTX PRO 4500 Blackwell GPU (32 GB VRAM). Llama Guard 3 8B was fine-tuned using Low-Rank Adaptation (LoRA) ($r = 32$, $\alpha = 64$, dropout 0.05, $\eta = 2 \times 10^{-4}$, 3 epochs, effective batch size 16, NF4 quantization during training, bfloat16 inference) on a custom WUT dataset (∼2800 samples). GPT OSS Safeguard was deployed in few-shot mode with a dedicated safety policy in the system prompt; the policy evolved iteratively from an initial zero-shot formulation, as zero-shot prompting proved insufficient to close attack vectors identified during red-teaming.

The taxonomy comprised 15 threat categories: S1–S14 from the MLCommons benchmark [10] plus the custom category S15 (Topic Restriction), flagging any query unrelated to the university.

The training set (∼2800 samples, 50/50 SAFE/UNSAFE) was generated using the PLLuM 8B Instruct model [14] with a role-assignment strategy – the model assumed the roles of a security auditor (generating UNSAFE queries for S1–S14), a red teaming expert (S15), and a WUT student (SAFE queries). A high sampling temperature ($T = 0.95$) was applied to ensure diversity. The synthetic test set comprised 2929 samples with uniform category representation.

A key contribution of this work is the creation of an independent ground truth dataset (1802 samples) serving as the primary test set. Queries were generated directly by over 30 WUT students during a 24-hour red teaming study, in which participants actively interacted with the Llama Guard 3-secured chatbot, attempting to trigger misclassifications. Collected queries were manually verified twice by the first author two weeks apart. Distribution: 58.9% SAFE, 41.1% UNSAFE; among UNSAFE, S15 (37.3%) and S13 (16.2%) dominated.

Evaluation was conducted at the binary level (Accuracy, Precision, Recall, F1, FP, FN) and the multiclass level (category assignment accuracy). Inter-model agreement and McNemar's test with continuity correction were also computed.

III. Results

A. Binary Classification

Table I presents results on the ground truth dataset.

TABLE I
BINARY CLASSIFICATION RESULTS ON GROUND TRUTH (n=1802) AND SYNTHETIC (n=2929) TEST SETS

	Llama Guard 3		GPT OSS Safeguard		TF-IDF
Metric	**GT**	**Synth.**	**GT***	**Synth.**	**GT**
Accuracy	69.0%	98.1%	77.8%	85.1%	56.6%
Precision	58.3%	97.3%	74.5%	93.4%	47.9%
Recall	85.5%	98.8%	69.1%	74.1%	64.7%
F1	69.4%	98.1%	71.7%	82.7%	55.1%
TP	633	1441	481	990	479
FP	452	40	165	70	521
TN	610	1431	849	1524	541
FN	107	17	215	345	261
FPR	42.6%	2.7%	16.3%	4.8%	49.1%

*92 samples without verdict (CoT token limit exceeded). †TF-IDF (char n-gram) + LogReg, trained on synthetic data only.

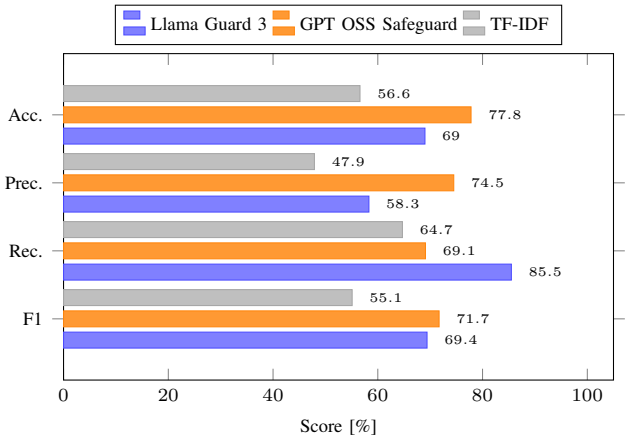


Fig. 1. Binary classification metrics on the ground truth test set (n=1802)

On the ground truth set, Fig. 1 and Table I reveal a fundamental trade-off: Llama Guard 3 exhibits high recall (85.5%), but at the cost of low precision (58.3%) – 452 FP indicate aggressive blocking of safe queries. GPT OSS Safeguard achieves higher precision (74.5%) and accuracy (77.8%), but misses more threats (215 FN). The FPR disparity – 42.6% for Llama Guard 3 versus 16.3% for GPT OSS Safeguard – directly quantifies this cost: in a real deployment, Llama

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 16:09:55 UTC from IEEE Xplore. Restrictions apply.

Guard 3 incorrectly blocks over two in five safe queries. McNemar's test – which assesses whether two classifiers differ significantly in their error rates on the same test set – confirmed statistical significance of the differences ($\chi^2 = 34.33$, $p < 0.001$), and inter-model agreement was only 63.2% (1139/1802), indicating substantial divergence between the two models.

On the synthetic set, both models achieved considerably higher scores. Llama Guard 3 results (F1=98.1%) should however be interpreted with caution – the model was fine-tuned on data generated by the same process as the test set, introducing a risk of performance overestimation. GPT OSS Safeguard shows a smaller synthetic-to-real gap (85.1%→71.7%, −13.4 pp) than Llama Guard 3 (98.1%→69.4%, −28.7 pp); this difference should be interpreted carefully, as the few-shot policy was iteratively refined with awareness of real query patterns from the red-teaming study, providing a slight distributional advantage. A key advantage of synthetic data is complete taxonomy coverage – the set contained uniform representation of all 15 categories, including S3, S4, and S8, which due to the academic context did not occur naturally during red teaming sessions. Synthetic data thus serves as a necessary complement for rare categories, but cannot replace evaluation on real-world data. For context, the TF-IDF (char n-gram) + Logistic Regression baseline trained exclusively on synthetic data (F1=55.1% on ground truth) confirms that LLM-based guardrails outperform this classical approach by 14–17 pp, demonstrating better generalization from synthetic to real user queries.

Analysis of ground truth errors reveals that Llama Guard 3's elevated FPR stems from two dominant patterns: over-triggering on topic restriction (S15) for short benign inputs such as greetings, and flagging hate speech (S9) when queries merely reference discriminatory language in an informational framing (e.g., "Can racist insults have legal consequences?", 100 cases). Both models share a systematic blind spot for authority-impersonation jailbreaks (S13) and academically-framed adversarial prompts (S2), which ranked among the most frequently missed threats in both classifiers (LG3: 53 FN, GPT OSS Safeguard: 59 FN).

B. Threat Category Classification

GPT OSS Safeguard demonstrated significantly higher category accuracy (74.4% vs 56.7%, $\chi^2 = 36.62$, $p < 0.001$). Fig. 2 shows the per-category breakdown.

Llama Guard 3 exhibits a strong bias towards S15 – 45 cases of S13→S15 confusion and 34 cases of S2→S15, stemming from the overrepresentation of this category in the training set (13:1 ratio). Table II lists the most frequent category confusions.

LoRA fine-tuning with S15 overrepresentation likely introduced a systematic classifier bias. GPT OSS Safeguard instead confuses S13 with S2 (14 cases) and S9 with S5 (10 cases), reflecting the objective difficulty of distinguishing these categories. Notably, the similar accuracy of both models on S15 (71.1% vs 78.5%) is architecturally interesting: Llama Guard 3

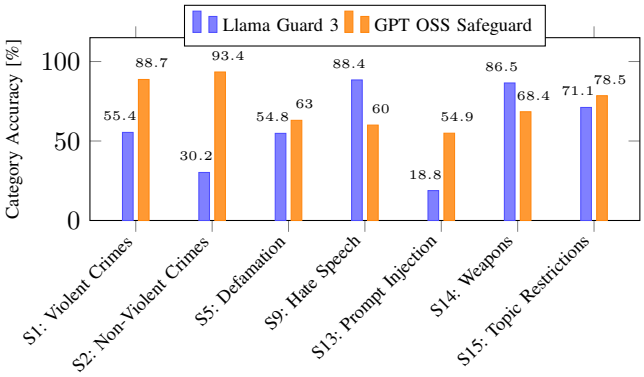


Fig. 2. Per-category threat classification accuracy (categories with ≥10 ground truth samples shown)

TABLE II
MOST FREQUENT CATEGORY CONFUSIONS (TOP 3 PER MODEL)

True → Predicted	Model	n
S13 → S15	Llama Guard 3	45
S2 → S15	Llama Guard 3	34
S15 → S9	Llama Guard 3	22
S13 → S15	GPT OSS Safeguard	16
S13 → S2	GPT OSS Safeguard	14
S9 → S5	GPT OSS Safeguard	10

achieves it through LoRA fine-tuning, GPT OSS Safeguard solely through the category description in the system prompt, suggesting that a precise domain policy description can partially compensate for the absence of specialized training. The threefold advantage of GPT in S13 attack detection (54.9% vs 18.8%) results from the multi-step intent decomposition in the CoT mechanism – Llama Guard 3 correctly blocked these attacks at the binary level, but misclassified them as S15, which is a diagnostic rather than a safety problem.

C. Impact of Quantization on Llama Guard 3

TABLE III
EFFECT OF QUANTIZATION ON LLAMA GUARD 3 ACCURACY AND MEMORY

Format	Acc.	F1	VRAM [MB]	Agr. w/ bf16
bfloat16	69.0%	69.4%	15 637	—
INT8	70.0%	69.8%	8 986	95.8%
INT4	65.6%	67.5%	5 759	90.5%

As shown in Tab. III, INT8 quantization represents the optimal trade-off: a 42.5% VRAM reduction (from ∼15.6 GB to ∼9.0 GB) with 95.8% binary classification agreement with the reference model and virtually identical F1 (69.8% vs 69.4%). A minimal Recall drop (84.3% vs 85.5%) with higher Precision (59.5% vs 58.3%) was observed, indicating a slightly less aggressive classifier. INT4 causes noticeable degradation: F1 drops to 67.5% (−1.9 pp.), and the model becomes more

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 16:09:55 UTC from IEEE Xplore. Restrictions apply.

aggressive (523 FP vs 452 in bfloat16), systematically over-classifying queries as S15.

D. Classification Latency

TABLE IV
CLASSIFICATION LATENCY COMPARISON OF GUARD MODELS

Metric	Llama Guard 3	GPT OSS Safeguard
Avg. time / sample	207 ms	10 575 ms
Throughput	4.8 samples/s	0.09 samples/s
Model size	8B (bfloat16)	20B

Llama Guard 3 is approximately $51\times$ faster (Tab. IV). The difference stems from the smaller model size and classification approach – GPT OSS Safeguard generates a multi-step CoT rationale ($\sim$500–1000 tokens) before issuing a label, whereas Llama Guard 3 generates only a label and category ($\sim$20 tokens). In a production deployment context, a latency of 10.6 s is unacceptable, while 207 ms is imperceptible to the user. Llama Guard 3 in INT8 configuration thus represents the optimal production choice: it operates locally without dependence on external APIs, and borderline queries could be delegated to the downstream model rather than unconditionally blocked. The system should also allow users to report unjustified blocks, providing data for iterative model improvement.

E. User Feedback

In the user study (over 30 WUT students), feedback was collected on Llama Guard 3 classifications. Students rated classifications as correct in 62.9% of cases and as incorrect in 20.9% (16.1% no response). Breakdown by prediction type reveals an asymmetry: of the 377 Incorrect responses, 238 (63.1%) concerned SAFE predictions and only 139 (36.9%) concerned UNSAFE predictions. Students more frequently disputed decisions to allow a query than to block it.

Comparison of expert labels with user perception revealed agreement of only 64.9% (948/1461). The asymmetry is clear: 465 queries labelled SAFE by experts were rated UNSAFE by students, while only 48 queries deemed UNSAFE were considered safe by students. Students systematically labelled a broader range of content as unsafe, including questions about alcohol, lecturers, and anti-discrimination topics in academic context. User perception thus proved more conservative than expert classification based on a precise taxonomy – implying that a ground truth dataset built solely on user ratings would overestimate the scope of threats. Evaluating guard models solely through technical metrics is insufficient in an educational context, where user experience and trust in the system are equally important.

IV. CONCLUSION

This paper presents one of the first comparative evaluations of LLM guard models in the context of a Polish-language academic chatbot. Three key conclusions are drawn.

First, the choice of guard model is a trade-off between safety and utility. Llama Guard 3 minimizes the risk of missing threats at the cost of excessive blocking; GPT OSS Safeguard minimizes disruption at the cost of higher risk.

Second, within the experimental context studied, open-source models showed competitive performance relative to commercial solutions for securing academic chatbots in under-represented languages. Llama Guard 3, with full data control and local deployment, suggests a viable path toward sovereign deployment in public institutions, though broader validation across institutions would be needed to generalise this conclusion.

Third, the low agreement between expert labels and user perception (64.9%) underscores that safety classification is inherently subjective and requires incorporating the end-user perspective.

The study has several limitations: the ground truth dataset comes from a single institution (WUT Faculty of Electrical Engineering students), categories S3, S4, and S8 did not appear in the collected data, and the evaluation did not cover structured adversarial attacks (GCG, AutoDAN). The shared training and test data distribution of the synthetic set artificially inflates Llama Guard 3 results on that set (F1=98.1% vs 69.0% on ground truth); the TF-IDF + LogReg baseline trained on the same synthetic data reaches F1=55.1% on ground truth, confirming the value added by the LLM-based approach.

Future work should cover: (1) systematic evaluation of robustness against structured adversarial attacks (GCG, AutoDAN) in a Polish-language context, (2) creation of a publicly available safety benchmark for Polish educational chatbots, (3) investigation of cascaded architectures in which a lightweight binary classifier performs initial filtering and a reasoning model analyzes only borderline cases, and (4) comparison with emerging dedicated Polish guard models such as Bielik Guard [13].

REFERENCES

[1] A. Wei, N. Haghtalab, and J. Steinhardt, "Jailbroken: How does llm safety training fail?" 2023, arXiv:2307.02483.

[2] A. Krasnodębska, K. Seweryn, S. Łukasik, and W. Kusa, "PL-guard: Benchmarking language model safety for Polish," in *Proc. Slavic NLP 2025*, Vienna, 2025, pp. 25–37.

[3] M. Zaharia *et al.*, "The shift from models to compound AI systems," *BAIR Blog*, Feb. 2024.

[4] T. Rebedea *et al.*, "NeMo guardrails: A toolkit for controllable and safe LLM applications," 2023, arXiv:2310.10501.

[5] P. Röttger *et al.*, "XSTest: A test suite for identifying exaggerated safety behaviours in large language models," 2023, arXiv:2308.01263.

[6] E. Kasneci *et al.*, "ChatGPT for good? On opportunities and challenges of large language models for education," *Learning and Individual Differences*, vol. 103, p. 102274, 2023.

[7] UNESCO, "Guidance for generative AI in education and research," UNESCO, Paris, France, Tech. Rep., 2023.

[8] A. Dubey *et al.*, "The Llama 3 herd of models," 2024, arXiv:2407.21783.

[9] H. Inan *et al.*, "Llama Guard: LLM-based input-output safeguard for human-AI conversations," 2023, arXiv:2312.06674.

[10] B. Vidgen *et al.*, "Introducing v0.5 of the AI safety benchmark from MLCommons," 2024, arXiv:2404.12241.

[11] OpenAI, "gpt-oss-safeguard technical report," Tech. Rep., Oct. 2025.

[12] E. Wallace *et al.*, "The instruction hierarchy: Training LLMs to prioritize privileged instructions," 2024, arXiv:2404.13208.

[13] SpeakLeash Consortium, "Bielik Guard: Efficient Polish language safety classifiers for LLM content moderation," 2026, arXiv:2602.07954.

[14] J. Kocoń *et al.*, "PLLuM: A family of Polish large language models," 2025, arXiv:2511.03823.

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 16:09:55 UTC from IEEE Xplore. Restrictions apply.